You are visitor No. 00

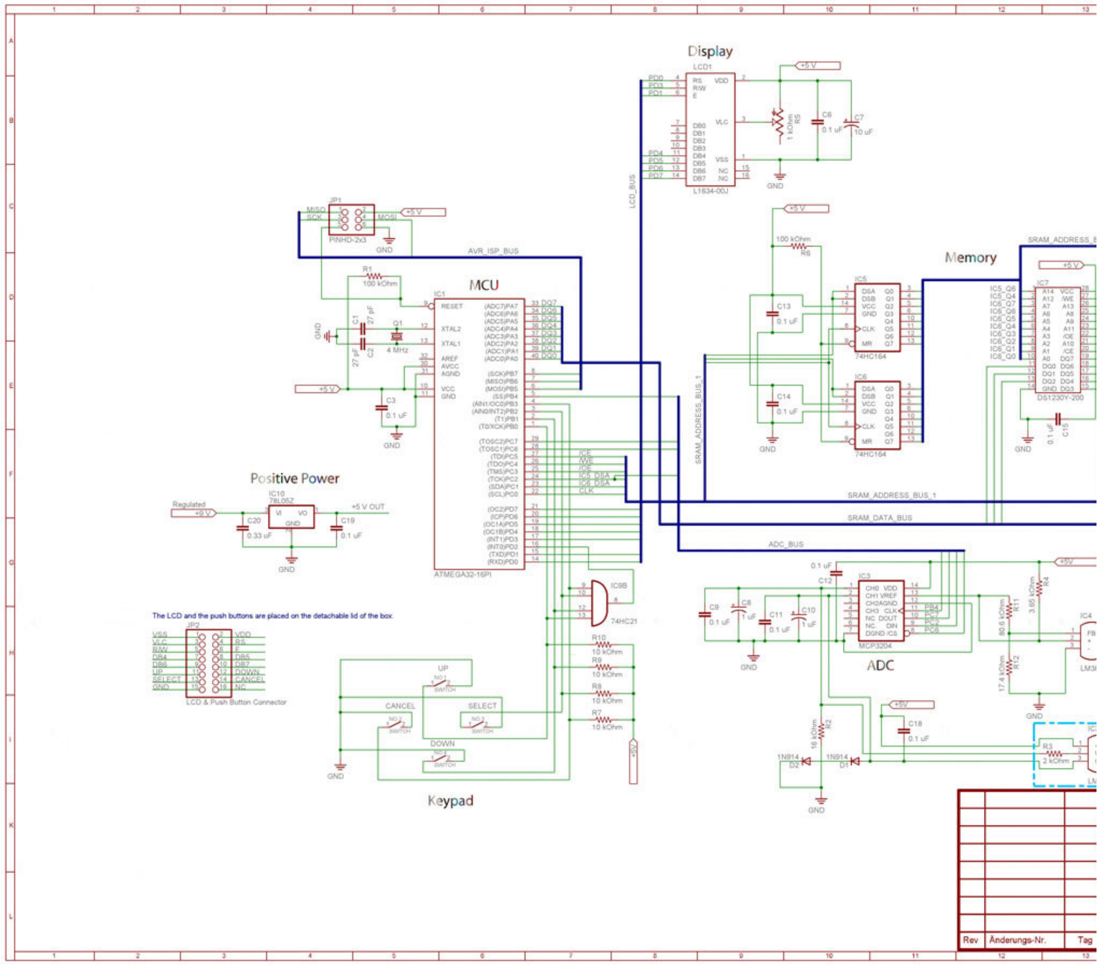
Incase you find the colors annoying there's a black and white version he

Introduction

I got tired of the poor memory functions of my thermometer so I built my ow
temperature on the thermometer I had but I wanted to know when those readi
it's own clock. In order to learn how to build a thermometer I had to learn abc
as it's called An Introduction to the Design of Small-scale Embedded Syste
Microcontrollers. Then I chose my microcontroller (the processor) to be an AV
embedded systems class at my university used AVR devices and the fact that
I live) ELFA didn't have any good development boards for PIC (The book focu
for Micro Controller Unit)). This turned out to be a very good choice. Becaus
can get lots of help with designing and implementing your system i.e. if you
programming. It's good to use the AVR-GCC compiler because there's a sp
GCC. There you can get even more help. I use the WinAVR package, wh
application Programmer's Notepad. I use Programmer's Notepad to program
system (external, because it's not situated on the STK500 development
manufactures all AVR devices, called ATSTK500 or just STK500. Debugging
emulator like the AVR-JTAG from Olimex Ltd. www.olimex.com or the more c
emulator. To be able to solder components, which are sensitive to electrosta
have the Temtronic WSD 81 from Weller. A lot of Integrated Circuits (ICs) a
advice to get things to work when designing an embedded system is to use
component. There's a lot of good information in the datasheets, so use them
300 pages long). For designing the schematic of my thermometer I used Eagle

The schematic

If you want to see a larger image of the schematic, Right-click on the sche
image of it. (Internet Explorer will not show images that large.)



Regarding the potentiometer for the LCD: The schematic is not wrong. The o
GND. It merely functions as a resistor, not as a voltage divider. Eagle also
automatically create a board layout in case you want to make a PCB layout (P
general purpose Sea-of-holes PCB and lots of wiring. Even though it's not in t
Power, MCU, LCD, Shift Registers, NV-SRAM, ADC, Sensor and Push Bu
consumption of each subsystem. Connect Vcc to the jumper and then take all
78L05 subsystem (positive power) I have it on the GND pin.) Below is listed th

Current consumption of the different subsystems

Positive Power	3.5 mA
MCU	7.0 mA
LCD	1.7 mA
Shift Registers	0.03 mA
NV-SRAM	0.05 mA

NV-SRAM	0.05 mA
ADC	0.5 mA
Sensor	0.1 mA
Push Buttons	1.5 mA

, which all together makes 21 mA. (???????) (I have a jumper on the whole s beginning of each minute, the current consumption drops to 14 mA, which is the lists may take several seconds. (app. 5 seconds) Normally it takes app. 0. sample the sensor 50 times. The ADC waits 5 milliseconds between each sa you go lower than that you're bound to get incorrect readings.

Try the thermometer

Below you can explore the functionality of the thermometer. Click on the tex alternatives of your choice, just as if you would have pushed up, or down to g cancel.

```
Date 2004-02-12
Time 08:55:14
  Temperature
    73,3°F
```

The real thermometer is controlled via 4 pushbuttons:

UP

CANCEL SELECT

DOWN

The cancel menu items don't exist on the real thermometer. Neither does th items. You press Cancel to get back. To get downwards in the menu-tree pres If the idle screen is shown and you push any button you arrive at the home m the end of a line indicates which line will be chosen when the SELECT butt since my LCD only had room for 8 user defined characters (It's not a cheap L

you traverse down the menu tree in the inline frame above and then click the c
you were before (that was too complicated)(If I have time in the future I'll make
to the menu where you were before, and the menu item which is marked as t
So if the idle screen is shown and you push a button you get to the home men
for its marked with real arrows). If you the choose Settings followed by Cance
marked since you chose Settings last. The different types of lists are explained

The different types of lists

`Today`

Today's highest reading between 00 am and 12 pm (or so far),
Today's lowest reading between 00 am and 12 pm (or so far) and
Today's lowest reading between 10 am and 6 pm (or so far) (Added since the l

`Today + the past 7 days`

Today + the past 7 days 8 highest readings between 00 am and 12 pm,
Today + the past 7 days 8 lowest readings between 00 am and 12 pm and
Today + the past 7 days 8 lowest readings between 10 am and 6 pm (Added s

`This month`

This month's 10 highest readings between 00 am and 12 pm (or so far),
This month's 10 lowest readings between 00 am and 12 pm and
This month's 10 lowest readings between 10 am and 6 pm

`2 months back`

The 59-62 highest readings between 00 am and 12 pm during the past 59-62 d
minimum number of days in 2 months is 59 and the maximum is 62),
The 59-62 lowest readings between 00 am and 12 pm during the past 59-62 d
The 59-62 lowest readings between 10 am and 6 pm during the past 59-62 da

If the list would be shorter than the number of days you would experience mar
warmer in July than in June the June readings will never enter the lists since th
May readings are out of date by the end of July (They are gradually phased ou

`6 months back`

The 181-184 highest readings between 00 am and 12 pm during the past 181-
December, the minimum number of days in 6 months is 181 and the maximum
The 181-184 lowest readings between 00 am and 12 pm during the past 181-1
The 181-184 lowest readings between 10 am and 6 pm during the past 181-18

`The past 12 months`

The 10 highest readings between 00 am and 12 pm for every month of the pas
The 10 lowest readings between 00 am and 12 pm for every month of the past
The 10 lowest readings between 10 am and 6 pm for every month of the past y

At 00:00:00 am the 1st of every month the lists of this month is moved to the p

`All Time`

The 100 highest readings between 00 am and 12 pm since the recordings star
The 100 lowest readings between 00 am and 12 pm since the recordings start
The 100 lowest readings between 10 am and 6 pm since the recordings starte

The long menus are setup in such a way the if you reach the bottom and pre top menu item is marked and at the bottom, while the bottom menu item is at rows and 16 characters for each row. In the set-time-and-date menu you decrease an entry (like hours e.g.) and you leave with the cancel button.

The different temperature scales

The thermometer can show the temperature including all of the lists in five diffe
Celsius (given the symbol c in the program code),
Fahrenheit (given the symbol f in the program code),
Kelvin (given the symbol k in the program code),
Rankine (given the symbol r in the program code) and
Réaumur (given the symbol é the program code since r was already busy).

Various Notes

Concerning the LCD & Push Button connector (situated on the lid of the euroc of the Push Buttons. So I only use 15 pins of my 2x8 connector. I could ha buttons and GND of the LCD, but I was already below 16.

The sensor-in-the-sun-function is advanced. It can even handle the thermome with the sensor being in the sun between e.g. 9pm and 3am between the 10th

XRAM is just an abbreviation for external SRAM.

This was the first and last project I develop without an emulator. The process c

I dropped it on the floor once but it still works.

You control the contrast of the display by turning the knob of the potentiometer

The sensor was prepared for outdoor use the following way:
First I soldered the wires, 3×2 meters of 55-408-03, to the sensor 73-088-10. 061-00. On top of that I put glass silicon to make it waterproof (be sure to get to perform well both in bright (no direct sunlight though) and dark conditions I at the bottom of the aluminum foil cover so water can seep out.

Connect the fuse in series between the +9V pin of the DC-Jack and the input on the OFF header the fuse is out of function.

Fuses and Lock Bits

The lock bits are set accordingly in the ATMega32-PI:
(The following are all checked. All others are not checked.)
Mode 1: No memory lock features enabled
Application Protection Mode 1: No lock on SPM and LPM in Application Sec
Boot Loader Protection Mode 1: No lock on SPM and LPM in Boot Loader S

The fuses are set accordingly in the ATMega32-PI:
(The following are all checked. All others are not checked.)
Boot Flash section size=2048 words Boot start address=$3800; [**BOOTSZ=00**
Brown-out detection level at VCC=2.7 V; [**BODLEVEL=1**]
CKOPT fuse (operation dependent of CKSEL fuses); [**CKOPT=0** (programme

to 1.)]
Ext. Crystal/Resonator High Freq.; Start-up time: 16K CK + 64 ms; [**CKSEL=1**

Pictures

Here are some pictures from the development process and the finished systen
(**Note I:** To download a picture right-click on it and choose "save target as")
(**Note II:** After I took the pictures of the cheap SRAM mounted I put all of the s holes for the pins unlike the bent tin sockets) so I wouldn't have to go throug right holes of the socket. After that I had one socket stacked on top of an oth when I put the NV-SRAM back in the socket it didn't fit as tight as it used to.)
(**Note III:** The different colors of the solder side pictures are explained below:
Red Vcc +5V
Black GND 0V
White Data to the LCD and to the shift registers
Brown Data I/O for the NV-SRAM
Orange address bus from the shift registers to the NV-SRAM
Purple push button output from the header to the AND gate to the MCU
Purple ISP bus
Green special. Separate lines, which didn't fit any other color.)

Software packages

And here's the software for the English version of the thermometer.
(**Note I:** I didn't find out until very late that you could use foe->xxx instead of *f
(**Note II:** The file tools contains a function ftoa, which converts floats = doubles
(**Note III:** It's 5'160 lines long or 178 kilobyte of c-code. It uses 28 kb of flash p

And here's the software for the Swedish version of the thermometer.
(**Note I:** I didn't find out until very late that you could use foe->xxx instead of *f
(**Note II:** The file tools contains a function ftoa, which converts floats = doubles
(**Note III:** It's 5'160 lines long or 178 kilobyte of c-code. It uses 28 kb of flash p

List of development tools used to develop the thermometer (Subject to an prices do not include VAT or the Swedish Moms. If you don't understand my ir the search box and you will get to the page of that special product. kr stands fo The XX-XXX-XX number is the ELFA part number. (e.g. 73-766-01)

1 pcs. 73-666-77 á 964:00 kr ATSTK500 starter kit
AVR-PG1B from www.olimex.com is a much cheaper alternative if you're just uses the standard 10 pin ISP-header (only 6 of them are used). In the sche (which you should be able to find in different places on the web) describes bo External Target System". (page 6-1) Just connect MOSI of the MCU to MOSI ISP10PIN header, SCK of the MCU to SCK of the ISP10PIN header, MISO (ISP10PIN header to Vcc of the thermometer and GND of the ISP10PIN heade headers:

ISP6PIN
  1 2
MISO VTG
SCK MOSI
RST GND

ISP10PIN
1 2
MOSI VTG
NC GND
RST GND
SCK GND
MISO GND

 pcs. 69-809-57 á 153:00 kr Power adaptor/Battery eliminator 12VDC/600mA
If you're not using the STK500 you just need the 9 V adaptor (69-808-17) usec
1 pcs. 82-888-88 á 79:20 kr LPM Solder sucker
1 pcs. 48-428-37 á 88:50 kr Breadboard Multiboard
1 pcs. 82-280-33 á 3217:00 kr ESD-proof soldering station Temtronic WSD 81
1 pcs. 82-254-84 á 49:50 kr Solder tip 2.0x0.8 mm for MLR-80, MPR-80 and W
(Better than the one you get with the soldering station I think.)
1 pcs. 80-861-59 á 49:00 kr Tiptinner/cleaner, type TTC 1 Vendor Multicore
1 pcs. 82-905-04 á 63:50 kr 0,7 mm 49 g 60% tin 40% lead solder 13.5 meters
1 pcs. 76-049-37 á 999:00 kr Multimeter Meterman 37XR, TRMS (With LCR-fu
1 pcs. 40-441-03 á 97:00 kr Black probeclip, 4 mm Vendor Hirschmann, type K
1 pcs. 40-441-29 á 97:00 kr Red probeclip, 4 mm Vendor Hirschmann, type Kl
1 pcs. 40-303-00 á 31:00 kr Labcord 4 mm Black Length 100 cm = 1 meter ap
1 pcs. 40-303-18 á 31:00 kr Labcord 4 mm Red Length 100 cm = 1 meter app.
1 pcs. Hot air gun á 279:00 kr + 4 mouthpieces and a case. Bought at my loca
store look for one in a hardware store. They are used to remove paint with in h
1 pcs. 43-782-53 á 5:72 kr Receptacle 1×2 contacts spacing 2,54 mm = 0.1" g
HV-100 (to measure the current consumption of each subsystem I drilled holes
and a red wire (with 7 connector strands) through each hole. De-isolated the w
them. To increase the strength I put 2.5 mm cord ends over the joint and crimp
on top of those and shrunk them with my hot-air gun. The heat shrink tubing sh
outer most cord ends. The other end of the red and black wires was then de-is
solder joints were isolated with heat shrink tubing. Just make shure you slip or
pins.)
1 pcs. 55-067-79 á 17:20 kr Heat shrink tubing FIT-221 polyolefin inner diamet

List of parts used for the thermometer (Subject to any changes in ELFA pa
at Stelios Cellar. The ELFA prices do not include VAT or the Swedish Moms. I
and enter e.g. 73-666-77 in the search box and you will get to the page o
currency in Sweden.)
The XX-XXX-XX number is the ELFA part number. (e.g. 73-766-01)

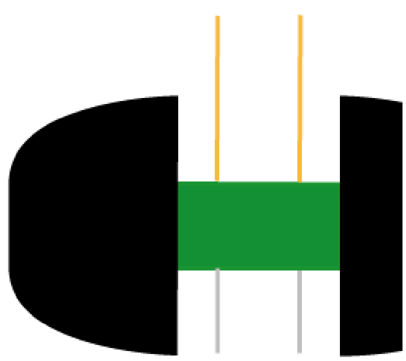
1 pcs. 73-766-01 á 261:00 kr DS1230Y-200 Sram with socket and battery 32K
1 pcs. 75-550-63 á 252:00 kr 16 characters by 4 rows alphanumeric LCD-displ
1 pcs. 73-088-10 á 79:10 kr LM 35C °Celsius temperature sensor Vendor Nati
1 pcs. 73-385-02 á 13:40 kr LM 385 adjustable voltage reference Vendor Natic
1 pcs. 74-515-29 á 22:90 kr Microprocessor crystal, HC49/4H, low profile, for p

55-254-07, 55-254-15, 55-254-23 etc. etc. Hook-up wire of as many colors as
0.25 mm^2 (should be about AWG 23) about 18 meters (60 feet) should be en
reel with 100 meters of hook-up wire. Your local supplier of electronics compo
matter. My local supplier charged me 1:50 kr per meter for any color.)

1 pcs. 67-008-01 á 1:63 kr 10 µF 50 V Electrolytic capacitor general purpose, '
2 pcs. 67-013-12 á 1:44 kr 1 µF 50 V Electrolytic capacitor general purpose, V
11 pcs. 65-656-59 á 2:23 kr 0,1 µF Ceramic plate capacitor Vendor Hitano Ra:
2 pcs. 65-848-74 á 00:873 kr 27 pF ceramic plate capacitor Vendor Hitano +-
less than more.) (called load capacitors)

4 pcs. 60-734-23 á 2:41 kr 10 kΩ Metal film restistors, 0,6 W Vendor Firstronic
2 pcs. 60-745-61 á 2:41 kr 100 kΩ Metal film restistors, 0,6 W Vendor Firstroni
1 pcs. 60-736-39 á 2:41 kr 16 kΩ Metal film restistor, 0,6 W Vendor Firstronics
1 pcs. 60-726-07 á 2:41 kr 2 kΩ Metal film restistor, 0,6 W Vendor Firstronics,
1 pcs. 60-729-12 á 2:41 kr 3,65 kΩ Metal film restistor, 0,6 W Vendor Firstroni
1 pcs. 60-736-70 á 2:41 kr 17,4 kΩ Metal film restistor, 0,6 W Vendor Firstroni
1 pcs. 60-744-54 á 2:41 kr 80,6 kΩ Metal film restistor, 0,6 W Vendor Firstroni

1 pcs. 48-137-54 á 14:00 kr 28 pin 0.6" wide DIP socket with turned pins Fabr
1 pcs. 48-138-12 á 19:60 kr 40 pin 0.6" wide DIP socket with turned pins Fabr
4 pcs. 48-135-64 á 7:12 kr 14 pin 0.3" wide DIP socket with turned pins Fabr
1 pcs. 48-135-49 á 4:42 kr 8 pin 0.3" wide DIP socket with turned pins Fabr S
1 pcs. 48-135-23 á 3:37 kr 6 pin 0.3" wide DIP socket with turned pins Fabr S
capacitors) Or you could just use all bent tin sockets like e.g. 48-162-03 if you
If you order 1 piece of 48-162-03 from ELFA you get 1 pipe of 17 sockets. Fo
a socket with turned legs/pins like 48-135-23
1 pcs. 73-420-58 á 48:30 kr MCP 3204 4-channel Serial 12-bit A/D with sampl
1 pcs. 48-320-10 á 62:90 kr 100×160 millimeters or any other Europa Card,
have solder islands and the hole spacing should be 2.54×2.54 mm = 0.1 inch
the cheapest Eurocards my local supplier had on the shelf. (In english boo
circuit boards.) I was not able to use outer most rows of holes. (see the picture
horisontal de-isolated wires are respectively the +5V line and the GND line. Ot
1 pcs. 64-414-30 á 4:13 kr 1,0 kΩ Trimpotentiometer Vendor Piher, type PT-6
1 pcs. 64-418-02 á 2:33 kr Axis for PT-6 Vendor Piher (turning knob)
You control the contrast of the display by turning the knob of the potentiometer
2 pcs. 43-717-79 á 10:60 kr Straight header spacing 2,54mm (0.1"), gold plate
0. You only need 2×5 or 2×3 depending on which ISP header you're using
consumption of each subsystem. You may damage some jumpers before you
the jaws on each side of the header and clip it off. (see picture below)



9 pcs. 43-710-43 á 2:73 kr Jumpers spacing 2,54 mm (0.1") Blue Vendor Perlc
1 pcs. 42-051-59 á 12:60 kr Adaptor connector (DC-jack) for circuit board mou
diameter

diameter
6 meters of 55-408-03 á 4:14 kr/meter 19 strands black hook-up w
HUBER+SUHNER type RADOX® 155 (to connect the sensor with the rest of t
2 pcs. 35-525-02 á 18:50 kr Black push button Vendor Miyama, series DS-663
1 pcs. 35-525-36 á 18:50 kr White push button Vendor Miyama, series DS-663
1 pcs. 35-525-10 á 18:50 kr Red push button Vendor Miyama, series DS-663.
1 pcs. 55-061-00 á 76:20 kr Heat shrink tubing FIT-300 melting inside 1.2 r
outdoor purposes or where good tightness is needed. Temperature range 110
1 pcs. 50-268-44 á 187:00 kr Instrumentbox for eurocard (100×160 mm cards)
2 pcs. 73-527-35 á 10:10 kr 74HC164 8-bit parallell-out serial shift register Ve
1 pcs. ATmega32-16PI á 164:00 kr (18:20 € from Stelios Cellar) DIL40 8-bit A
1 pcs. 69-808-17 á 97:00 kr Stabilized(regulatede output) battery-eliminator(ac

2 meter 55-264-21 red Hook-up wire. 7 condutor strands, cross section area a
(Note: If you order 55-264-21 from ELFA you get a reel with 100 meters of ho
may sell hook-up wire by the meter or foot for that matter. My local supplier ch

2 meter 55-264-21 black Hook-up wire. 7 condutor strands, cross section area

2 meter 55-264-21 white Hook-up wire. 7 condutor strands, cross section area

1 meter 55-264-21 green Hook-up wire. 7 condutor strands, cross section area
1 pcs. 73-508-12 á 6:42 kr 74HC21 Dual 4-input AND gate Vendor Philips Ser
15 pcs. 43-833-78 á 2:31 kr contact-elements for Ampmodu (for crimping. The
You can crimp them with a regular pair of flat pliers. I bought a pair of cheap c
cord ends were meant for 0.50 mm^2 wire, but I ended up not using tha sma
elements together first and then I crimped the isolated part of the wire and the
was in contact with the de-isolated part of the wire was full of solder and too ha
1 pcs. 43-835-01 á 24:50 kr Header for straight PC-mounting 2×8 pins for Amp
1 pcs. 43-833-11 á 5:02 kr Receptacle-house 2×8 pins for Ampmodu 0-02803
3 pcs. single á 5:00 kr Fast glassTubeFuse 50 mA 5×20 mm Vendor Littelfuse
1 pcs. 33-160-98 á 11:20 kr Fuseholder PTF/50 circuitboardholder 5×20 mm V
1 pcs. 73-001-07 á 4:12 kr Fixed positive voltage regulator 78L-series,
Semiconductor
1 pcs. 65-742-48 á 16:60 kr Ceramic plate capacitor 0,33 µF, multilayer Vend
2 pcs. 70-008-70 á 1:96 kr 1N914 Switchdiodes, low current Vendor Philips

Emacs

If you plan on using Emacs to edit your files you may want to use my .em
HOME, give it the value C:\home after first creating that directory. Put the fil
open a C-file the code will be displayed in different colors. Look in the .emacs-

Below follows a summary of the key-bindings of my .emacs-file:

Press ctrl-z to undo a command
(define-key global-map [(control z)] 'undo)

Press ctrl-f to open a file or just use the menus. If you want to create a new
havce to change the directory to what you want it to be. Otherwise the new f
the path of the file you can press tab to complete a name. (if you know there'
there are more files than one you will end up in a menu where you can choose
directory and press return. Remember Emacs was made long before Windov
wide spread back in 1975.)
(define-key global-map [(control f)] 'find-file)

Press ctrl-s to save a file (It prints out where it was saved.)
(define-key global-map [(control s)] 'save-buffer)

Put the cursor in the window you want to keep and press ctrl-w to delete other
corrupt .emacs-file) the window is split in two.
(define-key global-map "\^w" 'delete-other-windows)

Press esc-: to evaluate a LISP-expression (e.g. (global-set-key [(control c)] 'co

Press ctrl-g to cancel if you pressed e.g. ctrl-h and you want to stop

Press ctrl-h, [release and wait for emacs to print 'C-h'... at the bottom of the v
summary of all functions having to do with rectangles. This goes for all othe
paste is called kill, save and yank in Emacs-termonology.

Press ctrl-q to find and replace. You can use it just to find a certain sequenc
serch for, press ctrl-q, ctrl-y to paste, return, up-arrow (the four cursor keys)(t
a new instant of that string is found press: del to not replace, space to replace
can just type the word you want to search for.
(define-key global-map "\^q" 'query-replace)

You can cut, copy and paste vertically in Emacs
Use ctrl-d to copy a rectangle
(global-set-key '"\^d" 'save-rectangle)
Use ctrl-r to paste a rectangle
(global-set-key '"\^r" 'yank-rectangle)
Use ctrl-b to cut a rectangle
(global-set-key '"\^b" 'kill-rectangle)
I've even fixed it so that if you mark a rectangle, copy or cut that rectangl
rectangle, the marked rectangle is overwritten.

Windows style copy ctrl-c
(local-unset-key [(control c)])
(local-set-key [(control c)] 'copy-and-show) ; COPY-KEY.

Windows style cut ctrl-x
;;Comment out the two lines below if you need ctrl-x. (The character to comme
(local-unset-key [(control x)])
(local-set-key [(control x)] 'cut-and-show) ; CUT-KEY.

Windows style paste ctrl-v
(local-unset-key [(control v)])
(local-set-key [(control v)] 'paste-and-show) ; PASTE-KEY.

Mark a region or put the cursor on a specific line and press tab to indent respe
(define-key c-mode-base-map [(tab)] 'indent-funktionen)

Display files of different types in different colors so it's easier to program
(turn-on-font-lock)

Make the del-key and backspace-key work as in windows applications
(local-unset-key [DEL])
(local-set-key [DEL] 'backspace-funktionen)

Press alt-e to open the .emacs-file
(global-set-key [(meta e)] 'load-the-emacs-file)
The file has to here:
(find-file "c:/home/.emacs")

Paste in overwrite-mode
(global-set-key [(control t)] 'tie-yank-overwrite) ;

Press ctrl-left_windows_button (the one between ctrl and alt) to switch to the p
(define-key global-map [(control lwindow)] 'switch-to-previous-buffer)

Press ctrl-tab to switch to the next buffer among those, which are open.
(define-key global-map [(control tab)] 'bury-or-raise-buffer)

Here's an Emacs tutorial: http://www.lib.uchicago.edu/keith/tcl-course/emacs-t
Or if they move that page just search for Emacs tutorial on the web. Just rem
Emacs key-strokes wont work. To get back the original Emacs key-binding
the .emacs-file there or somewhere else where Emacs wont find it.

Background provided by Blue's Borders

The contents of this website may not be used for commercial p
Use the information at this website at your own risk. I'm not ta
cause you or your company.

I intentionally put everything on the same page so it
a string of text. Otherwise I would have to rely on
scanning truogh this site, which is not very likely.